

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

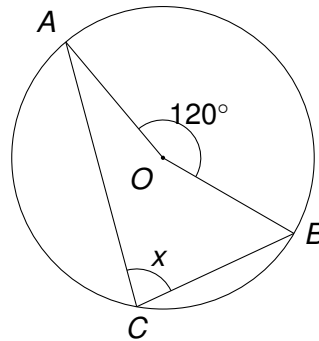
Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A , B and C are points on the circumference of a circle with centre O .

AOC passes through the centre such that O is the centre of the circle, and angle $AOB = 120^\circ$, where AOB is the angle at the centre.



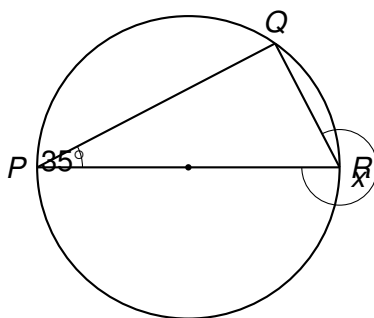
Find the size of angle x , the angle at the circumference.

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 1 is 2 marks)

- 2 A designer is creating a circular logo for an e-sports team. Points P , Q and R lie on the circumference of the circle, and PR is a diameter of the circle.



Angle $QPR = 35^\circ$.

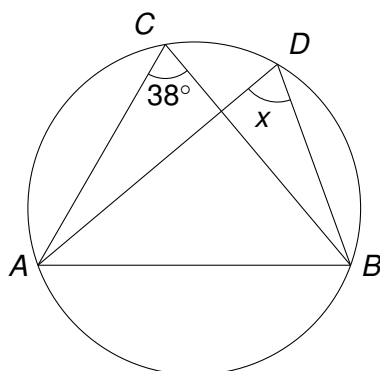
Work out the size of angle x , giving a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

- 3 Points A , B , C and D lie on the circumference of a circle, as shown in the diagram below. Chord AB divides the circle so that C and D lie on the same side of AB .



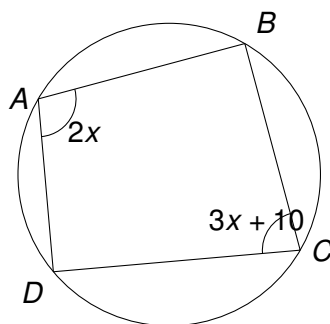
Angle $ACB = 38^\circ$.

Work out the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 3 is 3 marks)

- 4 A , B , C and D are four points on the circumference of a circle, forming a cyclic quadrilateral $ABCD$. Angle $A = 2x$ and angle $C = 3x + 10$, as shown in the diagram below.



- (a) Calculate the value of x .

$$x = \dots\dots\dots$$

(2)

- (b) Calculate the size of angle A and the size of angle C .

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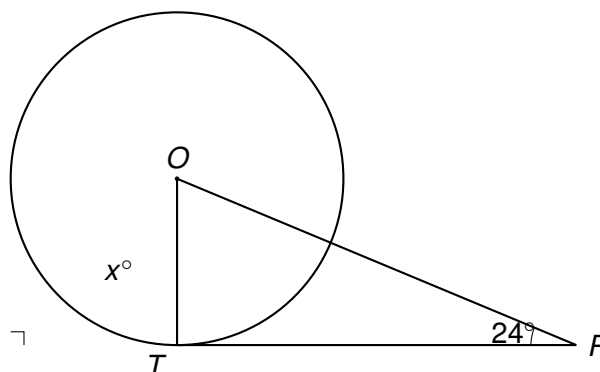
(2)

(Total for Question 4 is 4 marks)

- 5 An esports arena uses a large circular satellite dish, centre O , to receive a live broadcast signal for an international Valorant tournament. The signal travels along a straight line PT which touches the circular dish at point T , where PT is a tangent to the circle.

OP is a straight support cable from the centre O to a fixing point P outside the dish. The angle between the support cable OP and the tangent line OT extended...

More precisely: O is the centre of the dish, T is the point where the signal line touches the edge of the dish, and P is a point where the signal line meets a second support cable from O . Angle $OPT = 24^\circ$.



T is the point where the tangent line touches the circle, and O is the centre of the dish.
Angle $OPT = 24^\circ$.

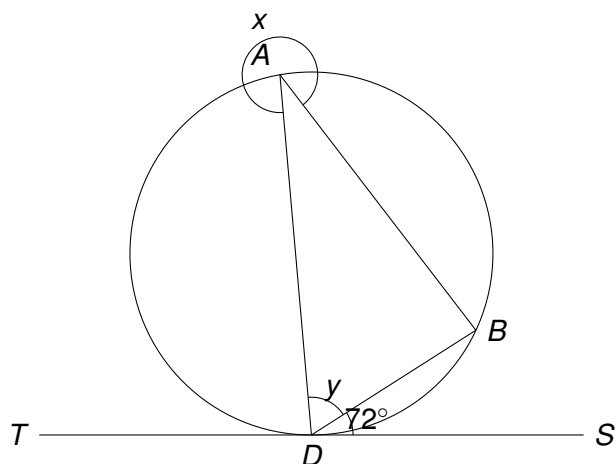
Work out the size of angle TOP , marked x° on the diagram. Give a reason for each stage of your working.

$$x = \dots\dots\dots^\circ$$

(4)

(Total for Question 5 is 4 marks)

- 6 A , B and D are points on the circumference of a circle. The line TDS is a tangent to the circle at D . Chord DB and chord DA are drawn, along with chord AB , as shown in the diagram.



The angle between the tangent TS and the chord DB is 72° , as shown.

- (a) Find the size of angle DAB , marked x on the diagram. Give a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(2)

- (b) The angle ADB (marked y on the diagram) is 53° .
Show that triangle ABD is **not** isosceles.

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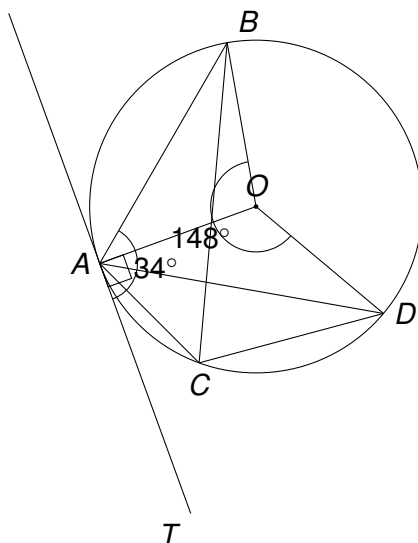
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(2)

(Total for Question 6 is 4 marks)

7 In the diagram, O is the centre of a circle. TA is a tangent to the circle at point A . B and C are points on the circumference, and $ABCT$ are positioned so that OAT is a straight tangent-radius configuration, with $ABCD$ a cyclic quadrilateral (where D lies on the circle such that OD is also a radius).

The angle between the tangent TA and the chord AB is 34° . The angle $BOD = 148^\circ$, where O is the centre and D is a point on the circumference such that OBD forms a triangle with two radii OB and OD .



(a) Triangle OBD is isosceles.

Show that angle $OBD = 16^\circ$, and state the reason why triangle OBD is isosceles.

(2)

(b) Calculate the size of angle BCD , giving reasons for your answer.

$BCD = \dots\dots\dots^\circ$

(3)

(Total for Question 7 is 5 marks)

TOTAL FOR PAPER IS ?? MARKS